About 2,600 results for “review wildfire spread prediction”

Fields of Study | Date Range | Has PDF | Author | Journals & Conferences | More Filters | Filters | Sort by Relevance

Facing the Wildfire Spread Risk Challenge: Where Are We Now and Where Are We Going?

Jingjing Sun · Wenwen Qi · Yuandong Huang · Chong Xu · Wentao Yang

Environmental Science · Fire · 7 June 2023

Wildfire is a sudden and highly destructive natural disaster that poses significant challenges in terms of response and rescue efforts. Influenced by factors such as climate, combustible materials,… Expand

14 PDF · PDF · Save · Cite

Advances in the prediction of spread through air spaces with imaging in lung cancer: a narrative review

Yun Wang · Deng-You Lyu · L. Fan · Shiyuan Liu · Medicine · Translational Cancer Research · 1 January 2023

TLDR STAS is an important invasion type of lung cancer, affecting the survival prognosis of patients, and computed tomography, 18F-FDG PET/CT, radiomics and deep learning are of great value in predicting STAS. Expand

4 PDF · PDF · Save · Cite

A new paradigm for forest fire spread prediction: Faster decisions at high resolution

I. González · C. Carrillo · A. Cortés · Antonio Espinosa · T. Margalef

Environmental Science, Computer Science · ICPP Workshops · 7 August 2023

TLDR The proposed algorithm, Boundary α -shape algorithm, has been design bearing in mind that it could be easily parallelized and highlights that the Boundaryα -shape allows a notable execution time reduction. Expand

ACM · Save · Cite

Forecasting Hourly Wildfire Risk: Enhancing Fire Danger Assessment using Numerical Weather Prediction

Christopher Rodell · Rosie Howard · P. Jain · N. Moisseeva · Timothy Chui · Roland Stull

Environmental Science · Weather and forecasting · 10 April 2024

Wildfire agencies use Fire Danger Rating Systems (FDRS) to deploy resources and issue public safety measures. The most widely used FDRS is the Canadian Fire Weather Index (FWI) System, which uses… Expand

3 · Publisher · Save · Cite

PyroScan: Wildfire Behavior Prediction System using Unmanned Aerial Vehicles (UAV)

Derek Thompson · Parker Padgett · Timothy Cary Johnson · Bhaskar Ghosh

Environmental Science, Engineering · 2024 IEEE World AI IoT Congress (AIIoT) · 29 May 2024

TLDR This work proposes a computer vision powered fire tracking and **prediction** application utilizing Unmanned Aerial Vehicles (UAV) to inform fire fighters regarding the probability of the direction of a fire. Expand

1 · IEEE · Save · Cite

Firecast Zigzag Convolutional Network for Wildfire Prediction

Yuzhou Chen · Joel Castillo · Huikyo Lee · Yulia R. Gel

Environmental Science, Computer Science · BigData Congress [Services Society] · 15 December 2024

TLDR The results indicate that the Firecast Zigzag Convolutional Network (F-ZCN) outperforms the current baseline methods for **wildfire prediction** and opens a path for more accurate **wildfire** risk analytics, even in scenarios of limited and noisy data records. Expand

IEEE Save Cite

Wildfire Autonomous Response and Prediction Using Cellular Automata (WARP-CA)

Abdelrahman Ramadan Environmental Science, Computer Science · arXiv.org · 2 July 2024

TLDR A novel approach that integrates terrain generation using Perlin noise with the dynamism of Cellular Automata (CA) to simulate **wildfire spread** and investigates emergent behaviors in MARL, focusing on efficient **wildfire** suppression and considering critical environmental factors like wind patterns and terrain features. Expand

[PDF] arXiv Save Cite

Predicting the rate of spread of mixed-fuel surface fires in northeastern China using the Rothermel wildfire behaviour model: a laboratory study

Hui Yang Huiying Cai Guang Yang Daotong Geng Long Sun Environmental Science · Journal of Forest Research · 4 July 2024

The rate of fire **spread** is a key indicator for assessing forest fire risk and developing fire management plans. The Rothermel model is the most widely used fire **spread** model, established through… Expand

3 Springer Nature Save Cite

Enhancing Alaskan wildfire prediction and carbon flux estimation: a two-stage deep learning approach within a process-based model

H. Seo Yeonjoo Kim Environmental Science, Computer Science · Environmental Research Letters · 28 October 2024

TLDR FireDL, a novel data-driven machine-learning model for the **prediction** of natural **wildfires**, is developed and combined with a land-surface model to better understand the impact of fire on carbon fluxes. Expand

2 PDF IOP Publishing Save Cite

Modeling Wildfire Spread with an Irregular Graph Network

Wenyu Jiang Fei Wang +5 authors Q. Meng Environmental Science, Computer Science · Fire · 4 November 2022

TLDR The results show that the IGN model can accurately and explicitly describe the spatiotemporal characteristics of the **wildfire spread** in a novel graph form while maintaining competitive simulation refinement and computational efficiency. Expand

19 PDF Publisher Save Cite

← … 6 7 8 9 … →

Stay Connected With Semantic Scholar

Your E-mail Address Sign Up

What Is Semantic Scholar?

Semantic Scholar is a free, AI-powered research tool for scientific literature, based at Ai2.

About
About Us
Publishers
Blog

Product
Product Overview
Semantic Reader
Scholar's Hub

API
API Overview
API Tutorials
API Documentation

Research
Publications
Research Careers
Resources

Help
FAQ
Librarians
Tutorials